The emergence of intelligent agents (1995 – present)

- One of the most important environments for intelligent agents is the Internet. AI systems have become more common in Web-based applications. Moreover, AI technologies underlie many Internet tools such as search engines, recommender systems, and Web site aggregators.
- The first conference on "Artificial General Intelligence (AGI)" was held in 2008. AGI looks for a universal algorithm for learning and acting in any environment.

10. The availability of very large data sets (2001 – present)

- More emphasis on data than algorithm.
- Increasing availability of very large data sources: for example, trillions of words of English and billions of images from the Web; billions of base pairs of genomic sequences.
- Today, many thousands of AI applications are deeply embedded in the infrastructure of every industry.

1.3 Different Types of AI/ Approach of AI

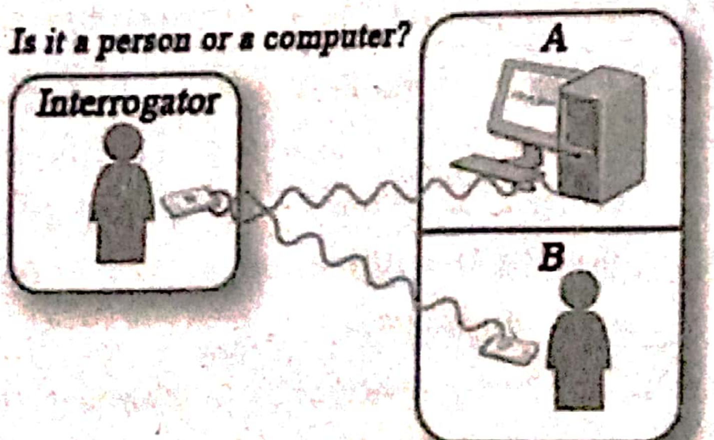
AI can be understood as the human-like intelligence exhibited by a machine. The field of AI is inter-disciplinary in which several sciences and profession converse such as computer science, psychology, linguistics, neuroscience etc. AI can be defined in terms of the following four categories i.e., four views.

i. Acting like human/acting humanly: Turing test approach

The art of creating machines that perform functions which require intelligence when performed by people is understood as acting like a human. This approach is also called a *Turing test*. Turing test, purposed by Alan Turing (1950), was designed to provide a satisfactory operational definition of intelligence. The Turing test measures the performance of intelligence machine against that of a human being.

The Turing test is a method for determining whether or not a computer is capable of thinking like a human. According to this test, a computer is deemed to have artificial intelligence if it can mimic human responses under specific conditions.

Consider the following scenario. There are two rooms A and B. One of the rooms contains a computer, and the other room contains a human. The interrogator is outside and does not know which room has a computer. He can ask questions through a teletype and receives answers from both rooms A and B. The interrogator needs to identify whether a human is in room A or in room B. To pass the Turing test, the computer has to fool the interrogator into believing that it is human.



The factors required to pass the Turing test are:

- **Natural language processing (NLP):** To enable it to communicate easier and successfully.
- **Knowledge representation:** To store information what it knows.
- **Automated reasoning:** To use the stored information to answer questions and to draw new conclusions.
- **Machine learning:** To adapt to new circumstances and to detect and extrapolate patterns.

ii. Thinking humanly: The cognitive modelling approach

This refers to the automation of activities that we associate with human thinking activities such as decision making, problem-solving, learning, etc. This phenomenon is also known as the *cognitive-based approach* in which the focus is not just on the behaviour and the input, but the output also looks at the reasoning process. Cognitive science brings together computer model forms AI and experimental technique from psychology to try to construct precise and testable theories of the working of the human mind.

This approach requires understanding how human thinks. A different way to determine how human thinks like experience, investigation, inspection, current perception etc.

E.g., General Problem Solver

iii. Thinking rationally: The "laws of thought" approach

The study of computations that make it possible to perceive reason and act is meant by thinking. This is also known as the *"laws of thought" approach*. These laws of thought were supposed to govern the operation of the mind; their study initiated the field called logic.

A system is said to be rational if it does the right thing for what it knows. It attempts to codify the normative premises in term of logic, deriving the result as a right thinking. Logic provides precise notations or statements about all kinds of things and relations between them.

E.g., For the premises, All men are students. Ram is a man. The result is: Ram is a student.

iv. **Acting rationally: The rational agent approach**

Computational intelligence is the study of the design of an intelligence agent. This model is also known as the rational agent approach. A *rational agent* is one that acts so to achieve the best outcome or when there is uncertainty, the best-expected outcome.

A rational agent is more general than the laws of thought approach, which emphasizes correct interference. Making correct interference is sometimes part of being a rational agent because one way to act rationally is to reason logically to the conclusion that a givers action will achieve one's goals and then act on that conclusion.

E.g., reflex agent.

1.4 AI and Related Fields

i. **Logical AI**

A program knows about the world, in general, the facts of the specific situation in which it must act, and its goals are all represented by sentences of some mathematical, logical language. The program decides what to do by inferring that certain actions are appropriate for achieving its goals.

ii. **Search**

AI programs often examine large numbers of possibilities, e.g. moves in a chess game or inferences by a theorem proving program. Discoveries are continually made about how to do this more efficiently in various domains.

iii. **Pattern recognition**

When a program makes observations of some kind, it is often programmed to compare what it sees with a pattern. For example, a vision program may try to match a pattern of eyes and a nose in a scene to find a face. More complex patterns, e.g. in a natural